Security incident report

Section 1: Identify the network protocol involved in the incident
The log shows DNS, TCP and HTTP. DNS finds the website's IP address, TCP establishes the connection, and HTTP requests the webpage. The browser connected first to yummyrecipesforme.com and later to greatrecipesforme.com. At 14:18:32, DNS returned 203.0.113.22 for yummyrecipesforme.com. An HTTP GET followed at 14:18:36. At 14:20:32, DNS returned 192.0.2.17 for greatrecipesforme.com. At 14:25:29, the browser established a TCP connection and sent an HTTP GET to that site.

Section 2: Document the incident
An attacker guessed the default administrator password using a brute force attack. They added JavaScript that prompted visitors to download a malicious file and changed the admin password. Running the file redirected visitors to a fake website. Customers reported the problem. Analysts confirmed it through sandbox testing, traffic analysis and inspection of the website code and downloaded file. The brute force attack is described in the scenario, but is not shown in the traffic log.

Section 3: Recommend one remediation for brute force attacks
The team should stop users from reusing old passwords and block default passwords. This reduces the risk of an attacker logging in with a password they already know. Passwords should also be changed if someone else may have learned them.

Another useful measure is two-factor authentication (2FA). Users need their password and an extra step, such as entering a one-time code from their phone. Even if an attacker guesses the password, they still need the code to access the account.